Marmara University – Faculty of Engineering – Department of Computer Engineering

Fall 2020 – CSE3055 Database Systems
Quiz #1

(05.11.2020 | Duration: 45 min)

1) [50 pts] Consider the following business rules for a *patient appointment system*:

- Each doctor has a physician id as unique identifier, and a name composed of first name and last name.
- Each patient has a patient id as unique identifier, a first name, a last name, and an address composed of address details, city and state.
- Each appointment has a date, a time (date and time both are as unique identifiers), a duration, and a reason. An appointment can have one or more reasons.
- Each bill has a bill number as unique identifier, an amount insured, an amount not insured, a date sent, and a status.
- Each payment has a receipt number as unique identifier, a payment amount, a date, a method, and overdue days calculated from date.
- Each insurance company has an insurance name as unique identifier, a benefits contact, a phone number, and a claims address.
- A doctor can be scheduled for many appointments, but may not have any scheduled appointment at all. Each appointment is scheduled with exactly one doctor.
- A patient can schedule one or more appointments. One appointment is scheduled with exactly one patient.
- An appointment must generate exactly one bill, and a bill is generated by only one appointment.
- One payment is applied to exactly one bill, and a bill can be paid off over time by several payments. A bill can be outstanding, having nothing yet paid on it at all.
- One patient can make many payments, but a single payment is made by only one patient.
- Some patients are insured by an insurance company. If they are insured, they can only carry insurance with one company. An insurance company can have many patients carry their policies.
- For patients that carry insurance, the insurance company will make payments, each single payment is made by exactly one insurance company.

Draw an ER diagram that captures the preceding information. Identify any constraints not captured by ER diagram.

2) [50 pts] Micro loans are small loans, which is beginning to gain popularity especially among borrowers in developing countries. The idea is to bring venture lenders together using information technology. Typically, the loans will be used to finance startup or development of the borrower's company, so that there is a realistic chance for repayment. The money in a loan can, unlike traditional loans, come from many lenders. In this problem, you must create an E-R model that describes the information necessary to manage micro loans. The following information form the basis for creating the model:

- Each borrower and lender must be registered with information about name and address.
- A loan starts with a loan request, which contains information about when the loan should at latest be granted, the total amount being discussed (US-dollars), and how long the payback period is. Also, a description is included of how the money will be used. The rent on the payment is calculated in the loan amount, which is to say, the full amount is not paid.
- Lenders can commit to an optional portion of the total amount of a loan request.
- When the commitments for the loan request covers the requested amount, the request is converted to a loan. If not enough commitments can be reached, the loan request is cancelled. A borrower can have more than one request, and more than one loan at a time, but can at most make one request per day.
- The loan is paid through an "intermediary", typically a local department of a charity, who has a name and an address.
- The borrower chooses when he or she will make a payment. Every payment must be registered in the database with an amount and a date (at most one payment per loan per day). The lenders share the repayment based on how large a part of the loan they are responsible for.
- If the loan is not repaid before the agreed upon deadline, a new date is agreed. The database must not delete the old deadline, but save the history (the deadline can be overridden multiple times).
- Each lender can for each borrower save a "trust", which is a number between 0 and 100 that determines the lender's evaluation of the risk of lending money to that person. The number must only be saved for the borrowers, for whom there has been made such an evaluation.

Create an EER diagram for the given information given above.

IMPORTANT NOTES:

1) Use blank A4 papers.

2) Write the following sentences on the first page of your document: "I hereby swear that the work done on this quiz is totally my own; and on my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for this quiz. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues." Write your full name and sign it.

3) In case of any form of copying and cheating on solutions, all parts will get ZERO points. You should submit your own work. In case of any forms of cheating or copying, both giver and receiver are equally culpable and suffer equal penalties. All types of plagiarism will result in zero points from the quiz.

4) Use your handwriting to draw the diagrams, your handwriting should be readable, clear and neat. Poor handwriting that cannot be read will not be graded.

5) Scan your work and merge all the pages as a single pdf file. Rename the filename as YourNumberQ1.pdf (e.g. 150119123Q1.pdf).

6) Submit your file to the site http://ues.marmara.edu.tr before deadline.

7) Do not send quiz submissions through e-mail or any other media. E-mail attachments or such will not be accepted as valid submissions.

8) You are responsible for making sure you are turning in the right file, and that it is not corrupted in anyway. We will not allow resubmissions if you turn in the wrong file, even if you can prove that you have not modified the file after the deadline.

9) No groups are allowed.

10) Grade evaluation may be done on selected parts of the quiz, so try to complete all parts of your quiz successfully.

11) No late submissions will be accepted.